Well Scale Simulator User Manual

March 28, 2026

Purpose

This document is a practical user manual for the MATLAB wellbore scaling simulator in this repository. It explains how the code is organized, how a simulation case is defined, how the main algorithm works, and where to find the outputs. The description follows the actual implementation in the repository and uses the Krafla KJ-9 case as the main example.

Repository Structure

The top-level layout is:

```
clean_code/
  main.m
  functions/
    init/
    Hydrodynamics/
    Chemistry/
    Wellspec/
    Graphics/
    IO/
  Simulation/
    Krafla/
      KJ-9/
  tools/
    result_view/
  docs/
```

The main roles of the code folders are:

- `functions/init`: read case files, create the state structure, initialize the grid, rock-temperature model, chemistry, graphics, and HDF5 output.
- `functions/Hydrodynamics`: solve the transient 1D flow equations for pressure, enthalpy, and mixture velocity.
- `functions/Chemistry`: advect chemical species, call PHREEQC, update mineral deposition or dissolution, and feed the changes back to geometry and roughness.
- `functions/Wellspec`: read casing and deviation files and convert them into interpolated well geometry.
- `functions/Graphics`: display well geometry, transient profiles, and wellhead time series during the run.
- `functions/IO`: save restart files, append HDF5 output, and load optional schedules such as `WHP.csv`.

- `tools/result_view`: standalone browser for the generated HDF5 results.

How to Run a Case

The entry point is `main.m`. The script adds the project folders to the MATLAB path, selects a simulation directory through `state.SimDir`, initializes the model, and starts the transient loop.

Typical usage:

```
cd /path/to/clean_code
matlab
>> main
```

Inside `main.m`, select the case by editing:

```
SimDir='Simulation/Krafla/KJ-9/';
```

During execution the code opens an interactive MATLAB UI with:

- live transient plots,
- wellhead diagnostics,
- pause and cancel buttons,
- PDF snapshot export,
- result save buttons,
- PHREEQC script export for the current chemistry state.

Simulation Case Folder

Each simulation case is defined by a folder under `Simulation/...`. For example:

```
Simulation/Krafla/KJ-9/
```

The most important input files are:

- `params.md`: numerical controls, boundary conditions, time settings, grid size, physical constants, chemistry flags, and save cadence.
- `CASTINGDETAIL.csv`: casing or liner geometry used to construct the well diameter and roughness profiles.
- `DEVIATIONDETAIL.csv`: well trajectory used to convert vertical depth to along-hole distance and slope.
- `Feedzones.csv`: feed-zone location and reservoir inflow properties.
- `chemistry.md`: aqueous and gas chemistry setup, kinetic minerals, and deposition parameters.
- `chemistry.pht`: PHREEQC template used to assemble the cell-wise geochemical problem.
- `Pressure.csv` and `Temperature.csv`: measured wellbore profiles used for comparison in plots.
- `InitTemperature.csv`: formation or initial temperature profile.
- `scale.csv`: optional observed scale thickness profile.
- `WHP.csv`: optional wellhead schedule. In the current implementation the pressure column is interpreted as gauge pressure in bar and converted internally to absolute pressure in Pa.

The required minimum setup is `params.md` plus the geometry files needed by `read_geom`. Practical cases usually also provide chemistry, feed zones, and reference profiles.

Main Algorithm

The algorithm description below is based on the implementation and on the numerical-method section in `docs/main.tex`.

1. Initialization

When `main.m` starts, the code:

1. reads the case folder and parses `params.md`;
2. loads well geometry, inclination, feed zones, measured profiles, and optional scale data;
3. builds the 1D grid along the well;
4. initializes diameter, roughness, rock-temperature fields, and chemistry state;
5. creates the graphics window and prepares output containers.

2. Steady initial condition

Before the transient run, the code constructs an initial flowing state using `initializeSteadyState`. Depending on the case settings, initialization can proceed top-down or bottom-up. This produces the initial profiles of pressure, enthalpy, and velocity used as the starting point of the transient simulation.

3. Hydrodynamic step

The transient hydrodynamic solver advances the 1D conservation equations for mass, momentum, and energy. The main unknowns are pressure, total enthalpy, and mixture velocity. Phase properties are reconstructed from steam-table relations and a drift-flux closure.

The solver is fully implicit. At each time step the nonlinear system is linearized by Newton iterations. The linearized block tridiagonal system is then solved with a matrix Thomas algorithm. This avoids strict CFL-limited time steps and is the core reason the model can be run for long transients with large physical contrasts.

4. Chemical transport and PHREEQC update

After the flow step, the code advances the transport of dissolved species using the updated phase fluxes. The chemistry module then passes pressure, temperature, composition, and phase-partition information to PHREEQC. PHREEQC returns updated fluid composition, mineral saturation indices, and reaction progress.

The coupling between transport and geochemistry is handled sequentially within each outer time step. Several Picard-style internal iterations may be used to stabilize the chemistry-transport update.

5. Scale growth and hydraulic feedback

Mineral precipitation or dissolution changes the scale inventory in each cell. The code converts mineral accumulation into:

- scale thickness,

- updated well diameter,
- updated wall roughness.

This is the main coupling mechanism in the simulator. As the diameter shrinks and roughness increases, friction and flow regime change, which in turn changes pressure, boiling depth, saturation state, and future deposition.

6. Output and restart

At the end of each time step, the code updates the on-screen plots and appends data to the HDF5 output file. At the end of the run it also saves a restart MAT file containing the current thermohydrodynamic and chemistry state.

Program Flow in `main.m`

The practical sequence in `main.m` is:

1. `initializeState`
2. `initializeGraphics`
3. `initializeSteadyState`
4. `initResultsH5`
5. optional `chemistryInitializeStateV2`
6. transient loop with `OneStep`
7. optional `chemistryStepV2`
8. append profile, chemistry, and wellhead results
9. save restart profile on completion

In the transient loop:

- `OneStep` advances the hydrodynamics;
- `updateTopPressureFromWHP` applies the current wellhead schedule if one is used;
- `chemistryStepV2` is called after the specified chemistry start time;
- `appendProfileH5`, `appendChemistryH5`, and `appendWellheadH5` save the results.

Output Files

The main output is an indexed HDF5 file created in the case directory, for example:

```
Simulation/Krafla/KJ-9/results_001.h5
```

The HDF5 structure contains:

- `/meta`: spatial coordinate and initial diameter profile;
- `/wellhead`: time series of flow rate, pressure, temperature, densities, quality, velocities, and outlet chemistry;
- `/chemistry`: chemistry-step diagnostics and calcium balance terms;
- `/profiles/<time>`: full spatial profiles at saved times, including pressure, enthalpy, temperature, velocities, densities, gas fraction, diameter, saturation indices, and element concentrations;

- /inputs: copies of params.md, chemistry.md, and chemistry.pht for reproducibility.

If save_csv = 1, the code also writes CSV profile snapshots under the case folder.

Additional outputs produced by the UI are:

- result_NNN.pdf: exported plot snapshots;
- results_NNN.mat: saved in-memory MATLAB state and history arrays;
- phreeqc_saved_<timestamp>.phr: last generated PHREEQC script;
- restart_profile_NNN.mat: restart file for continuing a calculation from the saved state.

Viewing Results

Use the standalone result browser from MATLAB:

```
addpath(fullfile(pwd, 'tools', 'result_view'));
result_view
```

You can also pass a case folder or a specific HDF5 file:

```
result_view('Simulation/Krafla/KJ-9/')
result_view('Simulation/Krafla/KJ-9/results_001.h5')
```

Typical Workflow for a New Case

1. Copy an existing case folder such as Simulation/Krafla/KJ-9/.
2. Edit params.md to set the grid, boundary conditions, run time, save cadence, and chemistry flags.
3. Replace geometry files, feed zones, measured profiles, and chemistry files with case-specific data.
4. Point main.m to the new SimDir.
5. Run main.m.
6. Inspect results_NNN.h5 with result_view.

Notes for Users

- The code works internally in SI units for the solver, but input pressure units may be specified in params.md through P_unit.
- Chemistry is optional and is activated with calc_chem = 1.
- The chemistry module starts only after stat_chem, which can be used to let the hydrodynamic state settle first.
- Scale affects both diameter and roughness, so hydraulic and geochemical changes should be interpreted together.
- Restart files preserve the current state of the flow and chemistry fields and are the correct starting point for continued long simulations.